1. **Draw factor trees and write the prime factorisation using index notation (e.g. $2^2 \times 3$):**

2. **Use the Venn diagram method to find HCF and LCM. Show all working:**

3. **For each word problem: choose HCF or LCM, explain why, then calculate:**

4. **The lesson says 1 is NOT a prime number because it has only ONE factor. Explain precisely why a prime number must have EXACTLY two factors. Write THREE sentences.**

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5. **A student says: 'To find the LCM of 12 and 18, I multiply them: $12 \times 18 = 216$.' Show the correct answer using the Venn diagram method. Then explain the student's mistake.**

My working:

The student's mistake was: ______________________________

6. The lesson says prime factorisation is like 'breaking a number into its atomic building blocks.' Use the example $60 = 2^2 \times 3 \times 5$ from the lesson to explain this analogy. Write FOUR sentences.